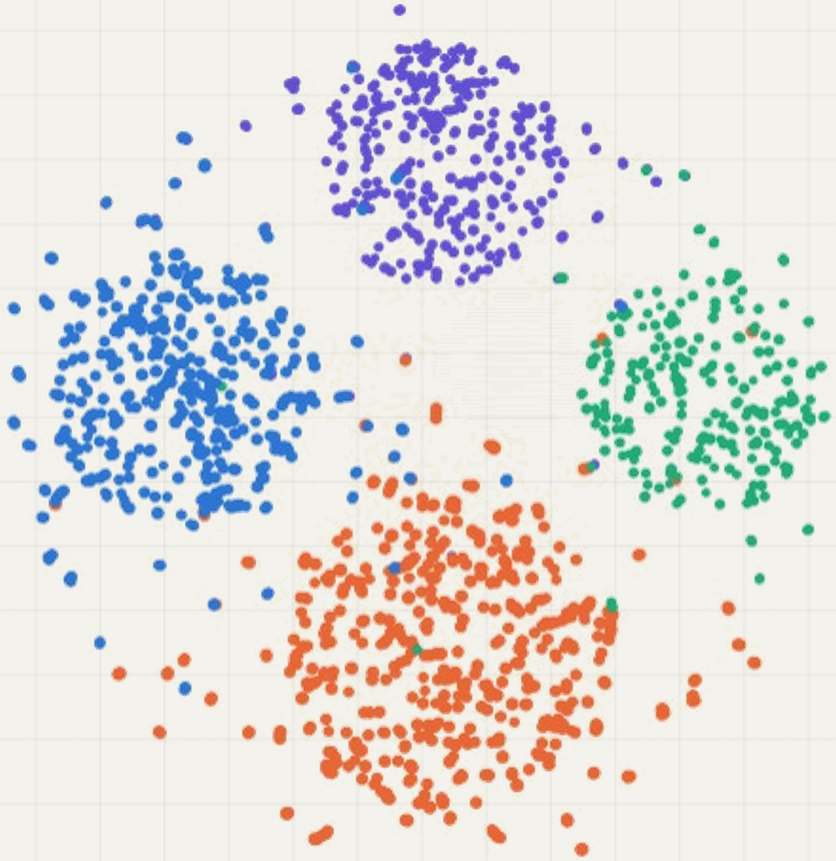
TEAM DUMBOCRACY

Clout.

An Election Game

NAME	ROLL NUMBER
Pakki Sai Kaushik	AI23BTECH11017
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Vishank Singh	AI23BTECH11032
B U Deekshith Patel	AI23BTECH11003



THE ELECTORATE · IN MATHS

How we build a world

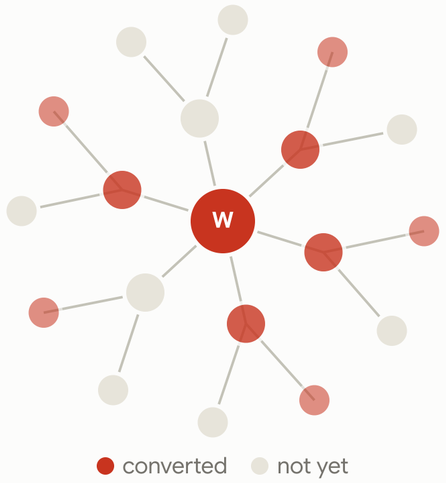
REGIONS	$n_r = m + \big[\mathrm{Mult}(N - 4m,\ \pi)\big]_r, \quad m = \lceil 0.15N \rceil$	Each region gets ≥ 15%; the rest split at random
FRIENDSHIPS	$P_{rr} = s\,p_r, \ \ P_{rs} = s\,p_{rs}, \quad p_r \sim U[0.5, 0.9],\ p_{rs} \sim U[0.01, 0.2]$	Stochastic block model; s sets mean degree 30
WANTS	$v_i(0) \sim \mathrm{Dir}\big(\kappa\,\mu_{\sigma(r)}\big), \quad \kappa = 60, \ \ v_i \in \Delta^5$	Budget split around the region's hidden archetype σ
MOVED BY	$\alpha_i,\ \beta_i,\ \gamma_i \sim \mathrm{Beta}\big(\tau(r)\big), \quad \Pr[\text{zealot}] = 0.05$	Friends, media, canvassing via hidden personality τ
DRIFT	$v_i \leftarrow (1 - a_i - c_i)\,v_i + a_i\,\bar{v}_{\mathcal{N}(i)} + c_i\,b, \quad a_i = 0.15\,\alpha_i$	Damped DeGroot step; c is the campaign pull
VOTES	$P_i = \mathrm{sig}\big(20\,(S_i - 0.825)\big), \quad S_i = 1 - \tfrac{1}{2}\lVert v_i - b \rVert_1$	Agreement with manifesto b sets the vote chance

TWO WAYS TO SPEND

Workers start cascades. Media gives a nudge.

Workers `20 coins`

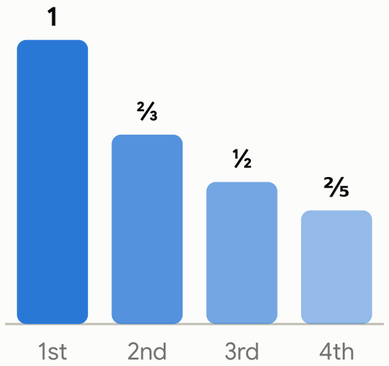
- A real voter, pinned to your manifesto
- Converts friends, **one hop per round**
- Hire early: more rounds to spread



Media `~50 coins`

- Reaches **60%** of a region at once
- A temporary pull, no network needed
- Each repeat airing is weaker

Strength ∝ `1 / (1 + 0.5·n)`



LINK The manifesto matters for both: voters who already like it convert more easily.

EPISTEMIC & TEMPORAL LOGIC

Knowing a win exists ≠ knowing how to win

The player can't see the world, so we track every world that **matches what they have seen**.

K true in **every** such world: the player *knows* it

M true in **some** world: it is *possible*

After each game, the debrief shows these verdicts for every round, alongside the hidden truth.

Knows *that*

every world has some winning plan

	π1	π2	π3
w1	1	0	0
w2	0	1	0
w3	0	0	1

No single plan is safe

Knows *how*

one plan wins in every world

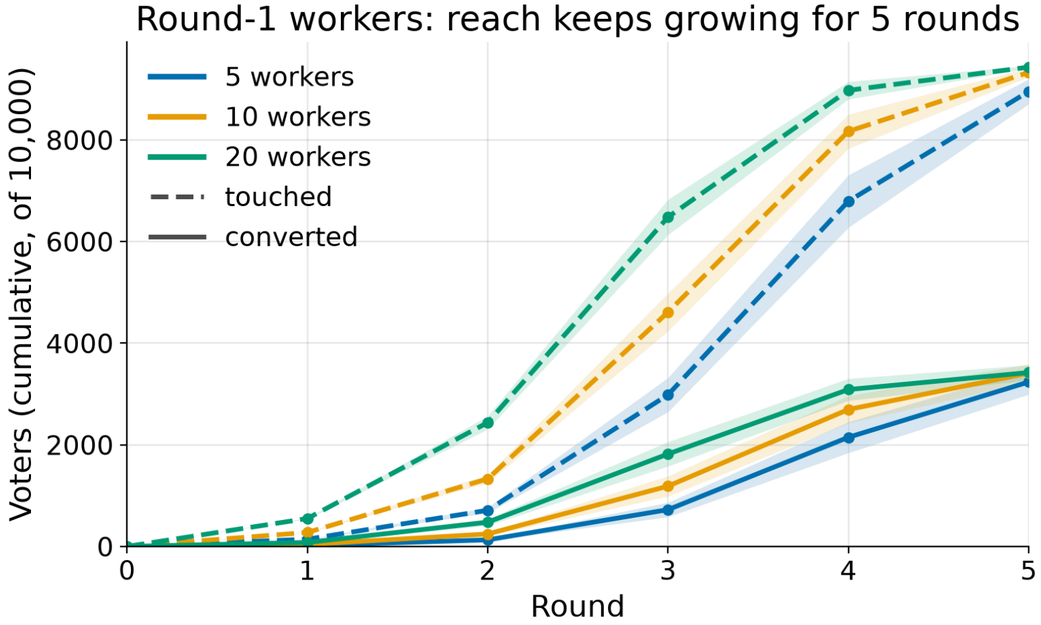
	π1	π2	π3
w1	1	1	0
w2	0	1	0
w3	0	1	1

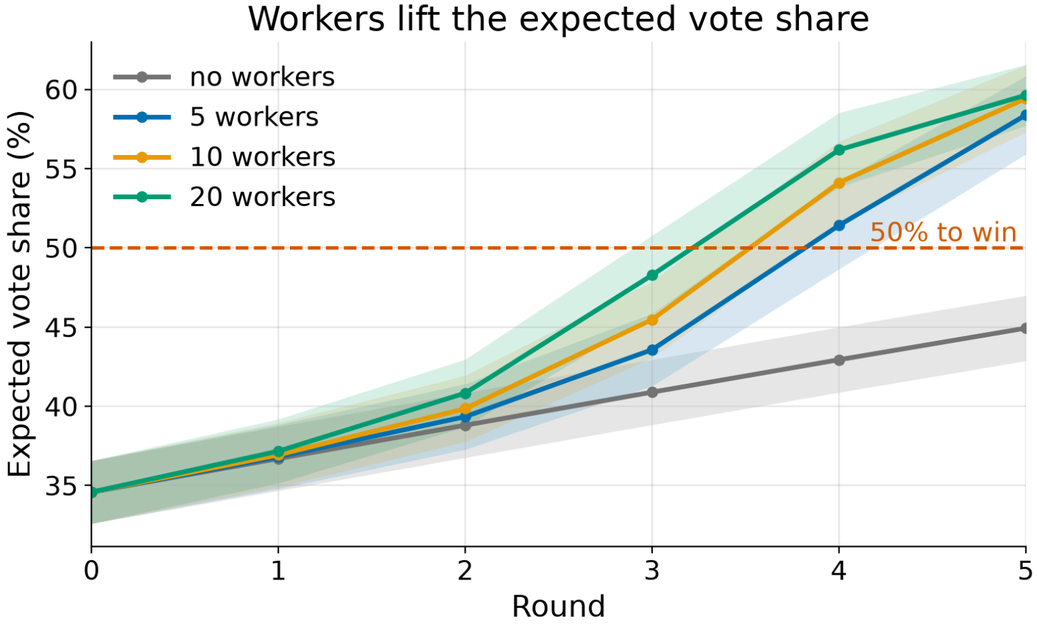
Play π2: guaranteed victory

Guaranteed ⇒ Knows a win exists ⇒ Possible

EXPERIMENT 1 · WORKERS BOUGHT IN ROUND 1 ONLY · 10,000 VOTERS

A few early workers reach almost everyone





45% (no workers) ⟶ 58% (5 workers) vote share:

The cascade touches ~9,000 voters and converts ~3,200.

More workers: sooner, not bigger

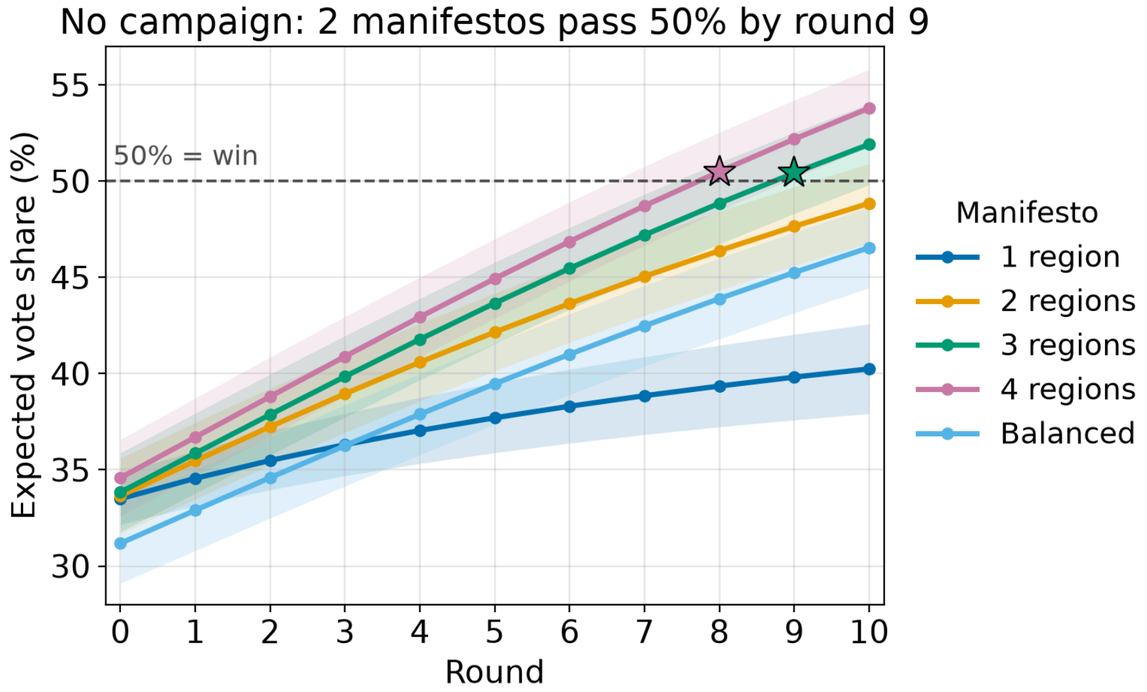
20 workers add only ~1 point over 5 by round 5; they mainly bring the gain forward.

The spread crosses regions

By round 5, about 30–37% of every region is converted.

EXPERIMENT 2 · FIXED MANIFESTOS, NO CAMPAIGN · 50 WORLDS

Blend more regions, win more votes

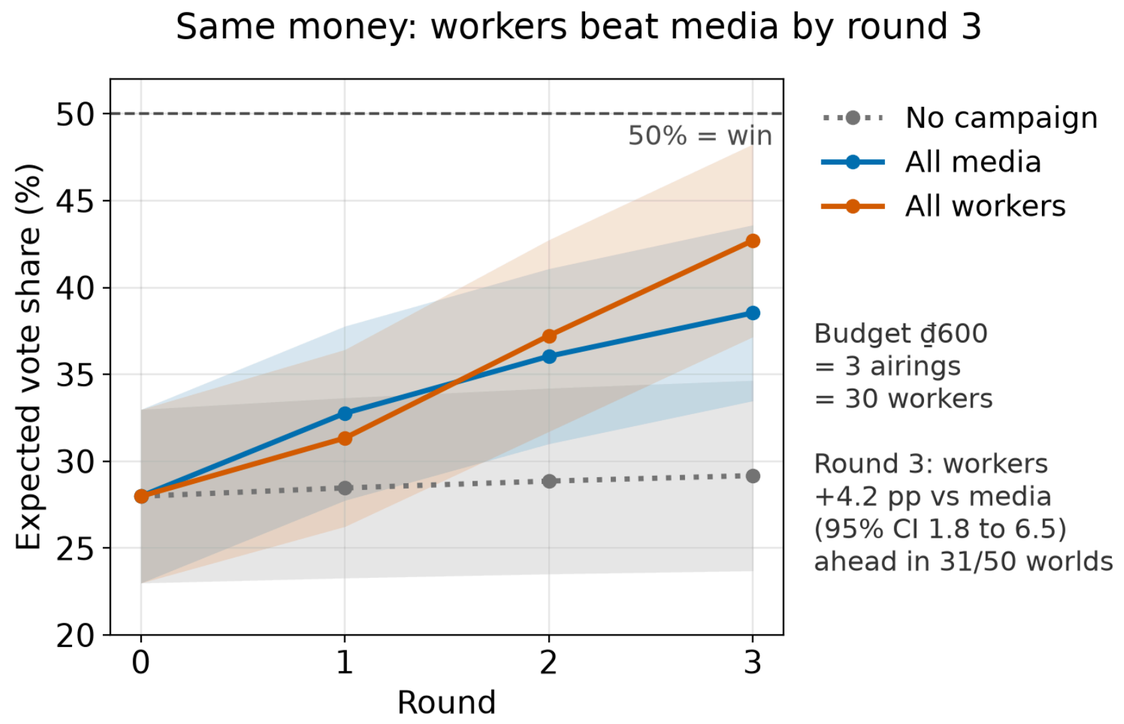


- **No manifesto wins alone within the round allocation for the game (5 rounds).** The blends do cross the threshold when extrapolated after the game duration.
- **The 4-region blend leads:** 44.9% at round 5 vs 37.3% for one region and when rolled out, it still performs better in a no-campaign environment.
- **A flat 20% split starts worst** (31%) and stays below every blend.

10,000 voters · no workers, media or surveys · manifestos built from archetype blends.

EXPERIMENT 3 · ALL MEDIA VS ALL WORKERS · SAME ₫600 · 50 WORLDS

Same money: workers overtake media



- **World building:** 2,500 voters in one region · 3 rounds · 20% manifesto · ₫600 = 3 airings = 30 round-1 workers.
- **Media leads after round 1,** but the worker cascade overtakes it from round 2.
- **Worker influence pulls ahead by round 3**: and they stay ahead in 30 of the 50 tested worlds.
- **Both investments beat doing nothing**.

Strategies

STRATEGY 1

Majority appeasement

The manifesto wins the big two; money wins the rest.

MANIFESTO	A size-weighted blend of the two largest regions.
SPENDING	Workers and media on the two smaller regions.
KEEP	A light nudge in the appeased regions against drift.

STRATEGY 2

Four-region blend

The manifesto reaches everyone; money locks in the big regions.

MANIFESTO	A size-weighted blend of all four regions.
SPENDING	Hyper-focus workers and media on the larger regions.

TEAM DUMBOCRACY

Individual contributions

G. Madhava Asrith Murthy AI23BTECH11007	Built the smart player (surveys, manifesto, planning) and ran the strategy experiments.
B. U. Deekshith Patel AI23BTECH11003	Built the electorate: regions, friendship network, preferences, zealots, drift and voting.
Vuppula Hemanth Reddy AI23BTECH11033	Designed what the player sees and buys: polls, surveys, workers, media and their prices.
Pakki Sai Kaushik AI23BTECH11017	Built logic specifications, the game engine and the GPU experiment harness.
Vishank Singh AI23BTECH11032	Built the web game: its screens, visualisations, debrief timeline and web API.

TEAM DUMBOCRACY

Thank you.

Questions?